Cloud Computing
Lab 5

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Section: F

1. Docker installation and running hello-world

```
PES1UG23AM343:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d
a6a
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs th
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent
it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/
```

2. Docker hello-world working wihout sudo

```
ln: failed to create symbolic link '/home/vardha/Downloads/Downloads': File exists
PES1UG23AM343:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

PES1UG23AM343:~$
```

3. Kubectl installed

```
[sudo] password for vardha:
PES1UG23AM343:~$ kubectl version --client
Client Version: v1.35.1
Kustomize Version: v5.7.1
PES1UG23AM343:~$ Z
```

4. Minikube download

```
l/bin/minikube
PES1UG23AM343:~$ minikube version
minikube version: v1.38.0
commit: de81223c61ab1bd97dcfcfa6d9d5c59e5da4a0cf
PES1UG23AM343:~$
```

5. Kubectl insralled

```
🏄  Done! kubectl is now configured to use "minikube" cluste
r and "default" namespace by default
PES1UG23AM343:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   8h    v1.35.0
PES1UG23AM343:~$
```

6. Namespace created

```
PES1UG23AM343:~$ kubectl create namespace pes1ug23am343-alph
a
namespace/pes1ug23am343-alpha created
PES1UG23AM343:~$ kubectl create namespace pes1ug23am343-beta
namespace/pes1ug23am343-beta created
PES1UG23AM343:~$ kubectl get namespaces
NAME                 STATUS   AGE
default              Active   12h
kube-node-lease      Active   12h
kube-public          Active   12h
kube-system          Active   12h
pes1ug23am343-alpha  Active   20s
pes1ug23am343-beta   Active   11s
PES1UG23AM343:~$
```

7. Namespace used

```
PES1UG23AM343:~$ kubectl config view --minify | grep namespa
ce:
    namespace: pes1ug23am343-alpha
PES1UG23AM343:~$ kubectl config view --minify | grep namespa
ce:
    namespace: pes1ug23am343-alpha
PES1UG23AM343:~$
```

8. Pod created and running

```
PES1UG23AM343:~$ kubectl run citizen --image=nginx --restart
=Never
pod/citizen created
PES1UG23AM343:~$ kubectl get pods
NAME       READY   STATUS              RESTARTS   AGE
citizen    0/1     ContainerCreating   0          5s
PES1UG23AM343:~$ kubectl get pods
NAME       READY   STATUS    RESTARTS   AGE
citizen    1/1     Running   0          22s
PES1UG23AM343:~$
```

9. Pod Description

```
PES1UG23AM343:~$ kubectl describe pod citizen
Name:             citizen
Namespace:        pes1ug23am343-alpha
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Tue, 17 Feb 2026 05:13:04 +0000
Labels:           run=citizen
Annotations:      <none>
Status:           Running
IP:               10.244.0.3
IPs:
  IP:  10.244.0.3
Containers:
  citizen:
    Container ID:   docker://2a231d75bc5b28b81df4250b730a4eb575abee2335516bc105a7d9aa2270c094
    Image:          nginx
    Image ID:       docker-pullable://nginx@sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e
262e0056d313208
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Tue, 17 Feb 2026 05:13:25 +0000
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-8sgzv (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-8sgzv:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
```

```
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-8sgzv:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  96s   default-scheduler  Successfully assigned pes1ug23am343-alpha/citizen to m
inikube
  Normal  Pulling    94s   kubelet            spec.containers{citizen}: Pulling image "nginx"
  Normal  Pulled     76s   kubelet            spec.containers{citizen}: Successfully pulled image "n
ginx" in 18.715s (18.715s including waiting). Image size: 160850673 bytes.
  Normal  Created    75s   kubelet            spec.containers{citizen}: Container created
  Normal  Started    75s   kubelet            spec.containers{citizen}: Container started
PES1UG23AM343:~$
```

10. Pod running

Terminal 1

```
PES1UG23AM343:~$ kubectl get pods -w
NAME      READY   STATUS    RESTARTS   AGE
citizen   1/1     Running   0          3m10s
temp      0/1     Pending   0          0s
temp      0/1     Pending   0          0s
temp      0/1     ContainerCreating   0          0s
temp      1/1     Running             0          5s
temp      1/1     Terminating         0          13s
temp      1/1     Terminating         0          13s
temp      0/1     Completed           0          15s
temp      0/1     Completed           0          15s
temp      0/1     Completed           0          16s
temp      0/1     Completed           0          16s
```

Terminal 2

```
ln: failed to create symbolic link '/home/vardha/Downloads/Downloads':
PES1UG23AM343:~$ kubectl run temp --image=nginx
pod/temp created
PES1UG23AM343:~$ kubectl delete pod temp
pod "temp" deleted from pes1ug23am343-alpha namespace
PES1UG23AM343:~$
```

11. Citizen logs

```
PES1UG23AM343:~$ kubectl logs citizen
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/02/17 05:13:25 [notice] 1#1: using the "epoll" event method
2026/02/17 05:13:25 [notice] 1#1: nginx/1.29.5
2026/02/17 05:13:25 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/02/17 05:13:25 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2026/02/17 05:13:25 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/02/17 05:13:25 [notice] 1#1: start worker processes
2026/02/17 05:13:25 [notice] 1#1: start worker process 29
2026/02/17 05:13:25 [notice] 1#1: start worker process 30
2026/02/17 05:13:25 [notice] 1#1: start worker process 31
2026/02/17 05:13:25 [notice] 1#1: start worker process 32
2026/02/17 05:13:25 [notice] 1#1: start worker process 33
2026/02/17 05:13:25 [notice] 1#1: start worker process 34
2026/02/17 05:13:25 [notice] 1#1: start worker process 35
2026/02/17 05:13:25 [notice] 1#1: start worker process 36
PES1UG23AM343:~$
```

12. Broken Logs

```
2026/02/17 05:13:25 [notice] 1#1: start worker process 35
2026/02/17 05:13:25 [notice] 1#1: start worker process 36
PES1UG23AM343:~$ kubectl run broken \
 --image=busybox \
 --restart=Never \
 --command -- sh -c "echo 'CRASH: something went wrong' >&2; exit 1"
pod/broken created
PES1UG23AM343:~$ kubectl logs broken
Error from server (BadRequest): container "broken" in pod "broken" is waiting to start: ContainerCre
ating
PES1UG23AM343:~$
```

13. Deployments

```
PES1UG23AM343:~$ kubectl create deployment pes1ug23am343 --image=nginx
deployment.apps/pes1ug23am343 created
PES1UG23AM343:~$ kubectl get deployments
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
pes1ug23am343   0/1     1            0           4s
PES1UG23AM343:~$ kubectl get pods
NAME                            READY   STATUS    RESTARTS   AGE
broken                          0/1     Error     0          61s
citizen                         1/1     Running   0          6m58s
pes1ug23am343-c6577c485-dbl27   1/1     Running   0          10s
PES1UG23AM343:~$
```

14. 5 deployment pods : scale up

```
pes1ug23am343   0/1     1            0           4s
PES1UG23AM343:~$ kubectl get pods
NAME                            READY   STATUS    RESTARTS   AGE
broken                          0/1     Error     0          61s
citizen                         1/1     Running   0          6m58s
pes1ug23am343-c6577c485-dbl27   1/1     Running   0          10s
PES1UG23AM343:~$ kubectl scale deployment pes1ug23am343 --replicas=5
deployment.apps/pes1ug23am343 scaled
PES1UG23AM343:~$ kubectl get pods
NAME                            READY   STATUS              RESTARTS   AGE
broken                          0/1     Error               0          101s
citizen                         1/1     Running             0          7m38s
pes1ug23am343-c6577c485-bn4hc   0/1     ContainerCreating   0          5s
pes1ug23am343-c6577c485-dbl27   1/1     Running             0          50s
pes1ug23am343-c6577c485-dc6db   0/1     ContainerCreating   0          5s
pes1ug23am343-c6577c485-jwd9d   0/1     ContainerCreating   0          5s
pes1ug23am343-c6577c485-trlx8   0/1     ContainerCreating   0          5s
PES1UG23AM343:~$
```

15. Scale Down

```
PES1UG23AM343:~$ kubectl scale deployment pes1ug23am343 --replicas=3
deployment.apps/pes1ug23am343 scaled
PES1UG23AM343:~$ kubectl get pods
NAME                            READY   STATUS    RESTARTS   AGE
broken                          0/1     Error     0          2m26s
citizen                         1/1     Running   0          8m23s
pes1ug23am343-c6577c485-bn4hc   1/1     Running   0          50s
pes1ug23am343-c6577c485-dbl27   1/1     Running   0          95s
pes1ug23am343-c6577c485-trlx8   1/1     Running   0          50s
PES1UG23AM343:~$
```

16. Rolling Update (Renovating City)

```
PES1UG23AM343:~$ kubectl get rs,po
NAME                                        DESIRED   CURRENT   READY   AGE
replicaset.apps/pes1ug23am343-c6577c485     3         3         3       2m15s

NAME                                READY   STATUS    RESTARTS   AGE
pod/broken                          0/1     Error     0          3m6s
pod/citizen                         1/1     Running   0          9m3s
pod/pes1ug23am343-c6577c485-bn4hc   1/1     Running   0          90s
pod/pes1ug23am343-c6577c485-dbl27   1/1     Running   0          2m15s
pod/pes1ug23am343-c6577c485-trlx8   1/1     Running   0          90s
PES1UG23AM343:~$
```

17. Update Image

```
PES1UG23AM343:~$ kubectl set image deployment/pes1ug23am343 nginx=nginx:1.25
deployment.apps/pes1ug23am343 image updated
PES1UG23AM343:~$ kubectl get rs,po
NAME                                       DESIRED   CURRENT   READY   AGE
replicaset.apps/pes1ug23am343-c6577c485    3         3         3       2m59s
replicaset.apps/pes1ug23am343-f64ccdfdf    1         1         0       3s

NAME                                 READY   STATUS              RESTARTS   AGE
pod/broken                           0/1     Error               0          3m50s
pod/citizen                          1/1     Running             0          9m47s
pod/pes1ug23am343-c6577c485-bn4hc    1/1     Running             0          2m14s
pod/pes1ug23am343-c6577c485-dbl27    1/1     Running             0          2m59s
pod/pes1ug23am343-c6577c485-trlx8    1/1     Running             0          2m14s
pod/pes1ug23am343-f64ccdfdf-tsnzx    0/1     ContainerCreating   0          3s
PES1UG23AM343:~$
```

18. Apply Service

```
PES1UG23AM343:~/PES1UG23AM343$ kubectl config set-context --
current --namespace=pes1ug23am343-beta
Context "minikube" modified.
PES1UG23AM343:~/PES1UG23AM343$ kubectl get services
NAME                    TYPE        CLUSTER-IP     EXTERNAL-
IP   PORT(S)   AGE
pes1ug23am343-service   ClusterIP   10.97.129.17   <none>
     80/TCP    3m19s
PES1UG23AM343:~/PES1UG23AM343$
```

19. Check Endpoints

```
PES1UG23AM343:~/PES1UG23AM343$ kubectl describe service pes1
ug23am343-service
Name:                     pes1ug23am343-service
Namespace:                pes1ug23am343-beta
Labels:                   <none>
Annotations:              <none>
Selector:                 app=nginx
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.97.129.17
IPs:                      10.97.129.17
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
Endpoints:                10.244.0.15:80,10.244.0.17:80,10.2
44.0.16:80
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>
```

20. Next step (traffic rerouting)

```
PES1UG23AM343:~/PES1UG23AM343$ kubectl delete pod -l app=ngi
nx
pod "nginx-pes1ug23am343-684598d475-nv5gk" deleted from pes1
ug23am343-beta namespace
pod "nginx-pes1ug23am343-684598d475-wpwwx" deleted from pes1
ug23am343-beta namespace
pod "nginx-pes1ug23am343-684598d475-z77jp" deleted from pes1
ug23am343-beta namespace
PES1UG23AM343:~/PES1UG23AM343$ kubectl get endpoints pes1ug2
3am343-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery
.k8s.io/v1 EndpointSlice
NAME                    ENDPOINTS
          AGE
pes1ug23am343-service   10.244.0.18:80,10.244.0.19:80,10.244
.0.20:80   4m48s
PES1UG23AM343:~/PES1UG23AM343$
```

21. Final step (browser)

```
PES1UG23AM343:~/PES1UG23AM343$ kubectl port-forward deployme
nt/nginx-pes1ug23am343 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
Handling connection for 8080
```

Hello from district PES1UG23AM343-beta running in Kubernetes